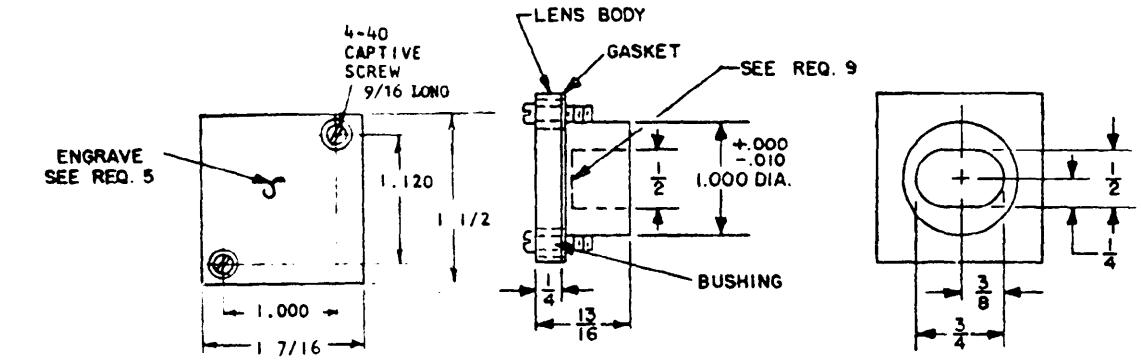
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MIL-L-3661/59A(SHIPS)
28 May 1971
SUPERSEDING
MIL-L-3661/59(SHIPS)
1 May 1968
MS17140A(SHIPS)(IN PART)
10 September 1964
(See page 2)

MILITARY SPECIFICATION SHEET

LENSES, INDICATOR LIGHT, STYLE LC45

The complete requirements for procuring the lenses described herein shall consist of this document and the latest issue of specification MIL-L-3661.



SH 9412A

	Inches	MM
(A)	0.010	0.25
	1/4	6.35
(A)	3/8	9.53
	1/2	12.70
(A)	9/16	14.29
	3/4	19.05
	13/16	20.64
	1.000	25.40
	1.120	28.45
	1-7/16	36.51
	1-1/2	38.10

Figure 1 - Style LC45 lenses.

(A) NOTES:
1. All dimensions are in inches. Unless otherwise specified, tolerances are ± 1/64 inch (0.38 mm) for fractions and ± 0.005 inch (0.13 mm) inch for decimals
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

(A) Denotes changes.

FSC 6210

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REQUIREMENTS:

1. Qualification: Not required.
2. Material:

(a) Lens body: Material for the lens body shall be as shown in table I and as follows:
(1) Red: Transparent, heat-resistant acrylic plastic having 90 ±5 percent light transmission in the range of 590 to 700 millimicrons wavelength.
(2) Blue: Transparent, heat-resistant acrylic plastic having 50 ±5 percent light transmission in the range of 425 to 475 millimicrons wavelength.
(3) Green: Transparent, heat-resistant acrylic plastic having 40 ±5 percent light transmission in the range of 475 to 550 millimicrons wavelength.
(4) Yellow: Transparent, heat-resistant acrylic plastic having 90 ±5 percent light transmission in the range of 525 to 700 millicrons wavelength.
(5) White: Translucent, heat-resistant acrylic plastic having 66.5 ±5 percent light transmission in 1/8 inch thickness.
(6) Colorless: Transparent, heat-resistant acrylic plastic having 95 ±5 percent light transmission in the range of 400 to 690 millimicron wavelength.
(b) Gasket: Rubber.
(c) Screw: Corrosion resistant steel.
(d) Bushing: Brass.

3. Design and construction: See figure 1.
4. Finish: All exterior surfaces, except the cavity on the back side of the lens shall be painted white 0.004 to 0.006 inch thick, followed by a coat of gray paint 0.002 to 0.005 inch thick. The paint on the outside of the lens shall be capable of withstanding the surface endurance test specified herein.
5. The face of the lens shall be engraved through the gray paint to allow either the color of the lens or the layer of white paint to appear.
6. Sealing: Dripproof.
7. Temperature cycling: Minus 55° to plus 125°C.
8. Shock: Method II.
9. No paint will be permitted on inside of cap.
10. Marking: The following markings shall appear on the back side of each lens:

(a) MIL-L-3661/59.
(b) Type designation.
(c) Manufacturer's name or trademark.

Table I - Style LC45 lenses.

Type designation 1/	Lens Material	Transmittance	Lens Color	Requirement
LC45RN3	Plastic	Transparent	Red	2(a)(1)
LC45BN3	Plastic	Transparent	Blue	2(a)(2)
LC45GN3	Plastic	Transparent	Green	2(a)(3)
LC45YN3	Plastic	Transparent	Yellow	2(a)(4)
LC45WT3	Plastic	Translucent	White	2(a)(5)
LC45CN3	Plastic	Transparent	Colorless	2(a)(6)

1/ Style LC45 lenses shall mate with any of the indicator light housings shown on MIL-L-3661/61 through MIL-L-3661/65.

Supersession data: This specification sheet replaces style 6 of MS17140.

QUALITY ASSURANCE PROVISIONS

First article inspection:
Complete lampholder assembly: When lenses specified herein are procured as a part of an assembly, the lenses with the indicator light housing shall be subjected to the applicable tests listed in the Qualification test of MIL-L-3661. The transmission and surface endurance tests specified herein shall also be conducted.

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Lenses: When lenses are procured separately, first article inspection shall consist of the following examination and tests:

Visual and dimensional examination	4.7.1
Sealing	4.7.11
Transmittance	(as specified herein)
Temperature cycling	4.7.5
Surface endurance	(as specified herein)

(A) Quality conformance inspection: when lenses are procured separately quality conformance inspection shall consist of the following:

Visual and dimensional examination
Sealing

Sampling for quality conformance inspection. A random sample of lenses shall be selected from each lot in accordance with MIL-STD-105, inspection level I. The AQL shall be 2.5 percent defective. Any lens in the sample containing one or more defects or failing to conform to the sealing test shall be rejected. If the number of nonconforming lenses in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

(A) Tests:

(A) Transmission. The transmission test to determine conformance with requirement 2(a) shall be conducted in accordance with MIL-C-25050.

(A) Surface endurance. The apparatus shall consist of a device which will apply a vertical load of 1.000 grams to the point of a needle held at an angle of 45 degrees to the horizontal. The apparatus shall hold the lens in a horizontal position and shall allow the entire length of the lens to pass horizontally under the needle in a rapid manner. The line scratched on the lens shall lie in the vertical plane containing the axis of the needle. The direction of movement of the lens relative to the needle shall be in the direction of the point. Figure 2 shows the general form of the equipment. The steel needle shall have a right conical point having a vertex angle of 16 to 20 degrees and a spherical tip with a radius of 0.002 to 0.0025 inch. A needle may be used for only one pass unless it is carefully checked to insure that its dimensions are within the limits indicated in Figure 2. The test shall consist of one pass of the needle on the front surface of the lens and one pass along the longest side of the lens, and shall not break the surface so as to cause light leakage from the lens or distortion of any marking.

Preparing activity:
Navy - SH
(Project 6210-N316)

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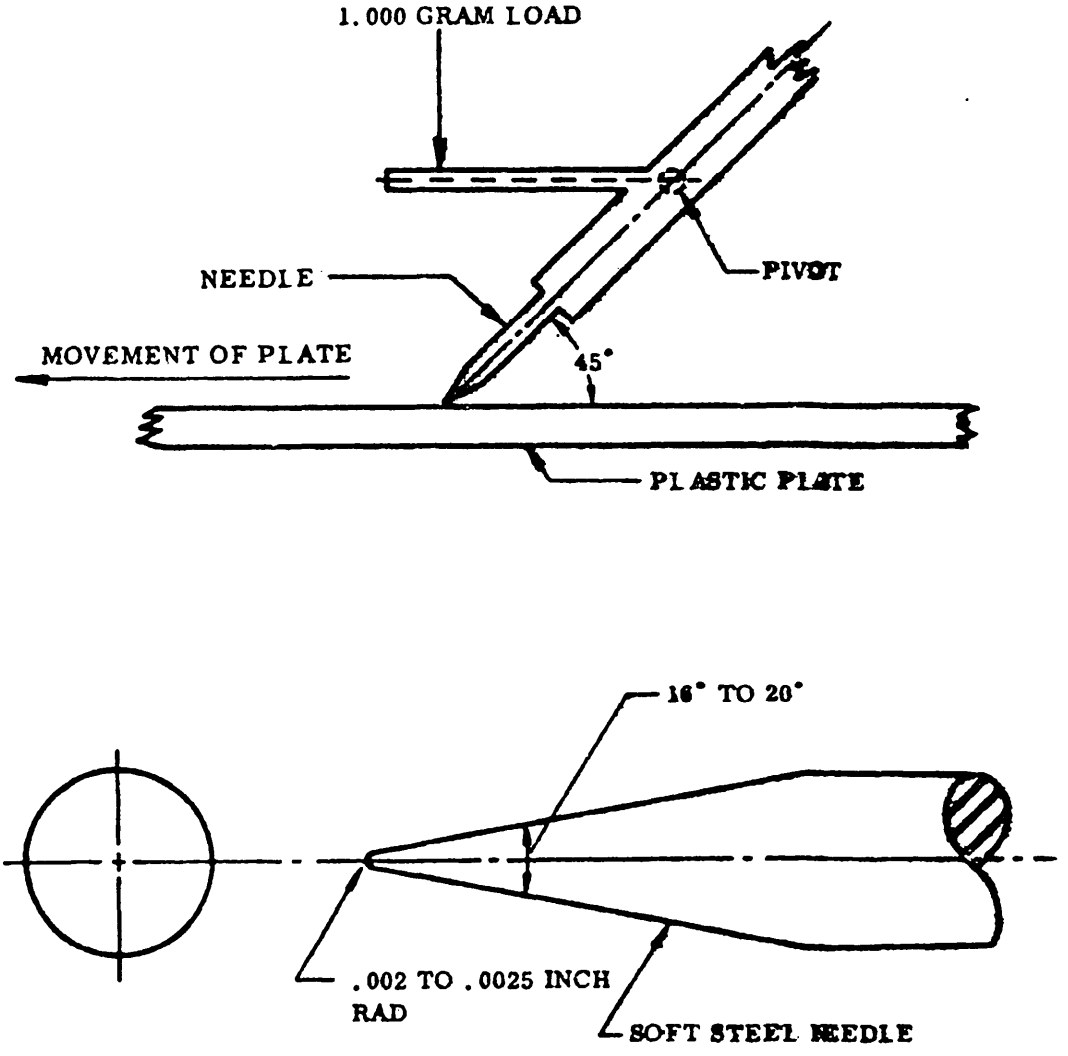


Figure 2 - Surface endurance test apparatus.

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